

CHAPTER 4 - Basic probability

SDS188

2026

1 General addition rule

1.1 Example 1

Use the contingency table to calculate the following probabilities:

	B	B'	Total
A	40	50	
A'	10	70	
Total			

1. $P(A)$

2. $P(A')$

3. $P(A \cap B)$

4. $P(A \cup B)$

1.2 Example 2

The cars have been categorised according to whether the car needs a warranty-based repair and the car's origin. Based on the data collected, the probability that the new car needs warranty repair is 0.08. The probability that the car was manufactured by a domestic company is 0.56 and the probability that the new car needs a warranty repair and was manufactured by a domestic company is 0.018. Construct a contingency table to evaluate the probabilities of a warranty-related repair.

Warranty-based repair	Domestic or foreign		Total
	Domestic	Foreign	
Yes	.	.	
No	.	.	
Total			

1. $P(\text{Yes})$

2. $P(\text{Yes} \cap \text{Domestic})$

3. $P(\text{Yes} \cup \text{Domestic})$

4. $P(\text{Yes} \cup \text{Foreign})$

2 Conditional probabilities

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

2.1 Example 3

	B	B'	Total
A	10	20	
A'	20	50	
Total			

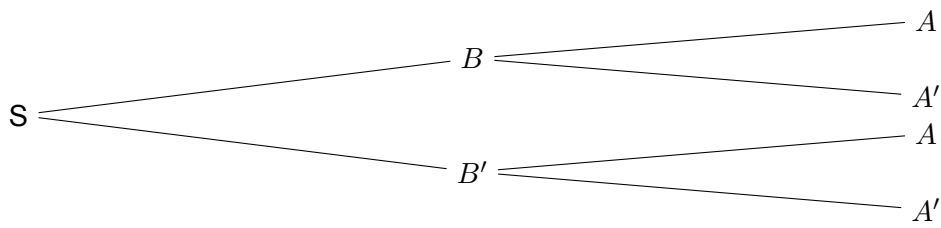
	B	B'	Total
A			
A'			1
Total			

1. $P(A|B)$

2. $P(A|B')$

3. $P(A'|B')$

4. Are events A and B independent?



2.2 Example 4

Suppose E and F are two events and that $P(E \cap F) = 0.2$ and $P(E) = 0.4$. What is $P(F|E)$?

3 Multiplication rule

$$P(A \cap B) = P(A|B)P(B)$$

OR

$$P(A \cap B) = P(B|A)P(A)$$

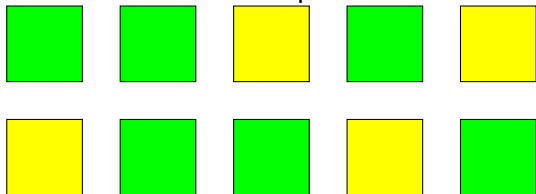
If independent $P(A \cap B) = P(A)P(B)$

3.1 Example 5

If $P(A) = 0.1$, $P(B) = 0.8$ and events A and B are independent, what is $P(A \cap B)$?

3.2 Example 6

A box contains 10 cell phones of which 4 are yellow and 6 are green.



1. If two cellphones are selected from the box without replacement, what is the probability that both phones are green?
2. What is the probability of selecting one yellow cellphone and one green cellphone?

3. If three cellphones are selected, what is the probability that all three cellphones are yellow?

4 Bayes Theorem

Calculate conditional probabilities given information regarding other relevant probabilities

Formulas:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_i)P(B_i) + \cdots + P(A|B_k)P(B_k)}$$

4.1 Example 7

Two cases (Simplest)

What is the probability of a hazardous fire when seeing smoke?

Information that is given:

- Hazardous fires are rare (1%)
- Smoke is common due to braai's (10%)
- 90% of hazardous fires cause smoke

$P(\text{Fire}|\text{Smoke}) =$

4.2 Example 8

Three cases

An art competition has entries of three painters: Pam, Pia en Pablo

- Pam has entered 15 paintings of which 4% has won first place
- Pia has entered 5 paintings of which 6% has won first place
- Pablo has entered 10 paintings of which 3% has won first place

What are the odds of Pam winning 1st place?

$P(\text{Pam}|\text{Eerste}) =$

4.3 Example 9

Lucretia tells the truth 2 out of 3 times. She tosses a die and says that it landed on a 4. What is the probability that it really landed on a 4?

5 Counting rules

Fundamentals of statistics - counting rules

Counting rule 1

k events

n trials

$$k^n$$

5.1 Example 10

Toss a coin 20 times

$k =$

$n =$

5.2 Problem 22 (from Tutorial 4)

If there are 10 multiple-choice questions on an exam, each having three possible answers, how many different sequences of correct answers are there?

$k =$

$n =$

Counting rule 2

$$(k_1)(k_2) \dots (k_n)$$

5.3 Problem 27

A particular brand of jeans is available in seven different sizes, three different colors and three different styles. How many different jeans does the store manager need to order to have one pair of each type in stock?

Counting rule 3

n is the number of items to be ordered

$$n! = (n) \times (n - 1) \times (n - 2) \times \cdots \times (1)$$

5.4 Problem 30

A gardener has six rows available in his vegetable garden to place tomatoes, eggplant, cucumbers, beans, onions, and lettuce. Each vegetable will be allowed only one row. How many different ways are there to position these vegetables in the garden?

Counting rule 4

X = items of interest

n = all items

$${}_nP_X = \frac{n!}{(n - X)!}$$

5.5 Problem 31

The Big Triple at a racetrack consists of picking the correct order of finish of the first three horses in the ninth race. If there are twelve horses entered in today's ninth race, how many Big Triple outcomes are there?

$n =$
 $X =$

Counting rule 5

X = items of interest

n = all items

$${}_nC_X = \binom{n}{X} = \frac{n!}{X!(n-X)!}$$

5.6 Problem 33

A student has seven books that she would like to place in a bag. However, only four books can fit into the bag. Regardless of arrangement, how many ways are there of placing four books into the bag?

$n =$
 $X =$